

HANNAH MULATO

Bioinformatics Software Engineer | Scientific Software | Genomics & Data Pipelines

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SUMMARY

Bioinformatics and Computational Biology graduate with 3+ years of software development experience building data-driven applications, APIs, databases, and computational workflows. Experienced in Python, Java, SQL, Linux/Bash, Git, and cloud/containerized environments, with hands-on project experience developing genomic analysis pipelines for WES/WGS data and integrating biological resources into research software. Background spans bioinformatics, backend/full-stack development, data processing, and technical troubleshooting, with a strong interest in scalable and reproducible software for genomics and life-science applications.

TECHNICAL SKILLS

Bioinformatics & Genomics: WES/WGS analysis, NGS workflows, variant calling & annotation, genomic data processing, biological databases, sequence analysis, computational biology

Programming & Data: Python, R, SQL, Java, C++, Pandas, NumPy, SciPy, Matplotlib

Software Engineering: FastAPI, Spring Boot, REST APIs, GraphQL, React, Node.js, JavaScript, TypeScript, JSON, data integration

Databases: PostgreSQL, MySQL, MongoDB, AWS DynamoDB

Scientific Computing & Infrastructure: Linux, Bash, Snakemake, SLURM, Conda, Docker, Kubernetes, AWS, Azure

Development Tools: Git, GitHub, GitLab, Jenkins, GitHub Actions, CI/CD, Agile/SCRUM

SELECTED BIOINFORMATICS PROJECTS

HPC Genomics Analysis Pipeline | *Python, Snakemake, SLURM, Linux, BWA-MEM, GATK, SAMtools/BCFtools*

- Developed an end-to-end WES/WGS workflow to automate genomic sequence processing from FASTQ alignment through BAM processing, variant calling, VCF generation, and functional annotation.
- Implemented reproducible orchestration with Snakemake and Conda, with SLURM-based execution to support parallel processing across samples and genomic regions.
- Integrated established genomic tools and downstream processing to summarize variant distributions and functional impacts, emphasizing reproducibility, modularity, and scalable execution.

Genelnsight - Bioinformatics Research Platform | *Python, FastAPI, React, REST APIs, UniProt, PubMed, AlphaFold, Ensembl*

- Co-developed a full-stack bioinformatics platform that consolidated biological and research data from multiple external resources into a unified application.
- Built and integrated Python/FastAPI endpoints for retrieving, validating, transforming, and serving biological data; implemented JSON mappings, caching, asynchronous requests, and error handling.
- Contributed features for protein structure exploration, gene information, scientific literature retrieval, and CRISPR guide analysis within a four-person development team.

PROFESSIONAL EXPERIENCE

Junior JavaScript Developer | MentorAPM

Aug 2023 - Dec 2024

Germantown, MD

- Developed and maintained React, GraphQL, and PostgreSQL features for data-visualization applications used to interpret and monitor complex technical data.
- Supported data retrieval, validation, and organization across MongoDB, MySQL, and PostgreSQL; optimized queries and application workflows to improve performance.
- Collaborated with developers, product stakeholders, and technical users to troubleshoot issues, clarify requirements, document system behavior, and support testing and iterative releases.
- Worked with AWS, Azure, Docker, and Kubernetes environments and contributed across development, testing, deployment support, maintenance, and product improvements.

Full-Stack Developer | Cognizant / Revature

Jun 2021 - Nov 2022

Germantown, MD

- Developed and tested RESTful APIs, Java/Spring Boot services, React interfaces, and database-backed applications supporting data handling, retrieval, and reporting.
- Applied SQL and NoSQL database design and optimization techniques to improve query performance and system reliability.
- Created technical documentation and troubleshoot backend services, application workflows, and database interactions while translating user requirements into technical solutions.
- Used Git/GitHub, Jenkins, and AWS-based tooling to support version control, automated builds, testing, deployment pipelines, and software quality practices.

EDUCATION

B.S., Bioinformatics & Computational Biology | University of Maryland, Baltimore County (UMBC) | May 2026

Relevant coursework: Genomic Data Analysis, Statistical Methods for Bioinformatics, Molecular Biology & Genetics, Database Management & SQL for Life Sciences, Bioinformatics Programming

Certificate, Full-Stack Development | Thinkful | 2021